



1. What are Numbers & Strings?

Numbers store numeric values. Strings store text values. They are two of the most common data types in JavaScript.

25

number

"Hello"

string



2. Why They Matter

- Store ages, prices, and scores.
- Store names, messages, and text.
- Build dynamic output.
- Support calculations and formatting.

3. Number

123

Used for
numeric values.

```
let age = 25;
```

4. String

"

Used for
text values.

```
let name =  
  "Ayla";
```

5. typeof

typeof

Checks a value's
data type.

```
typeof 25  
// number
```

6. Integer / Float



JavaScript uses the
same number type for
whole numbers and
decimals.

10 19.99

7. Quick Example



A number and a string
are different types.

10 !== "10"





8. Template Literals

Use backticks to insert variables into text.

```
let name = "Ayla";  
let age = 25;  
console.log(`Hello ${name},  
you are ${age}`);
```



9. String Concatenation

Combine text using the + operator.

```
let city = "Baku";  
console.log("City: " + city);
```



10. length

Counts characters in a string.

```
"JavaScript".length  
// 10
```



11. Case Methods

Change letters to upper or lower case.

```
text.toUpperCase()  
text.toLowerCase()
```



12. trim()

Removes extra spaces from the start and end.

```
"  Ayla  ".trim()  
// "Ayla"
```





13. includes()

Checks whether a string contains text.

```
message.includes("JS")
```



14. slice()

Extracts part of a string.

```
"JavaScript".slice(0, 4)  
// "Java"
```



15. replace()

Replaces one part of a string with another.

```
"I like Java".replace("Java",  
"JavaScript")
```



16. Chaining Methods

Methods can work together in one expression.

```
username.trim().toUpperCase()
```

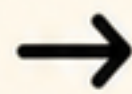
Result: clean and formatted text.



1. Type Conversion

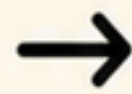
Change one data type into another.

String



Number

Number



String



2. Number()

Converts a string into a number.

```
Number("25")  
// 25
```



3. parseInt / parseFloat

Convert text to integer or decimal numbers.

```
parseInt("12.8") // 12  
parseFloat("12.8") // 12.8
```



4. toFixed()

Controls decimal places.

```
let price = 19.567;  
price.toFixed(2)  
// "19.57"
```



toFixed() returns a string.



1. Math Methods

Useful helpers for working with numbers.

```
Math.round(4.6) → 5
```

```
Math.floor(4.9) → 4
```

```
Math.ceil(4.1) → 5
```



2. Implicit Conversion

JavaScript sometimes changes types automatically.

```
"5" + 2 // "52"
```

```
"5" - 2 // 3
```



3. Quick Example

Build a readable message with a template literal.

```
const name = "Ayla";  
const score = 95;  
console.log(`Student: ${name},  
Score: ${score}`);
```

Output: Student: Ayla, Score: 95



4. Common Mistakes

- Forgetting to convert input values
- Confusing number and string types
- Using + and expecting math with a string
- Ignoring extra spaces in text



Small type mistakes can change your result.



1. Practice

Try these steps yourself:

- Create name and age variables.
- Convert age from string to number.
- Trim a username string.
- Print a message with a template literal.

```
let name = "Ayla";  
let age = "25";
```



2. Challenge Answer

If the code is:

```
let username = " ayla ";  
let result = username.trim()  
.toUpperCase();
```

- result -> "AYLA"



3. Mini Project

Build a simple User Profile Generator.

```
const name = "Ayla";  
const age = Number("25");  
const city = "Baku";  
let status = age >= 18 ? "Adult" : "Minor";  
console.log(`${name} from ${city} is ${status}`);
```

Output: Ayla from Baku is Adult



4. Key Takeaway

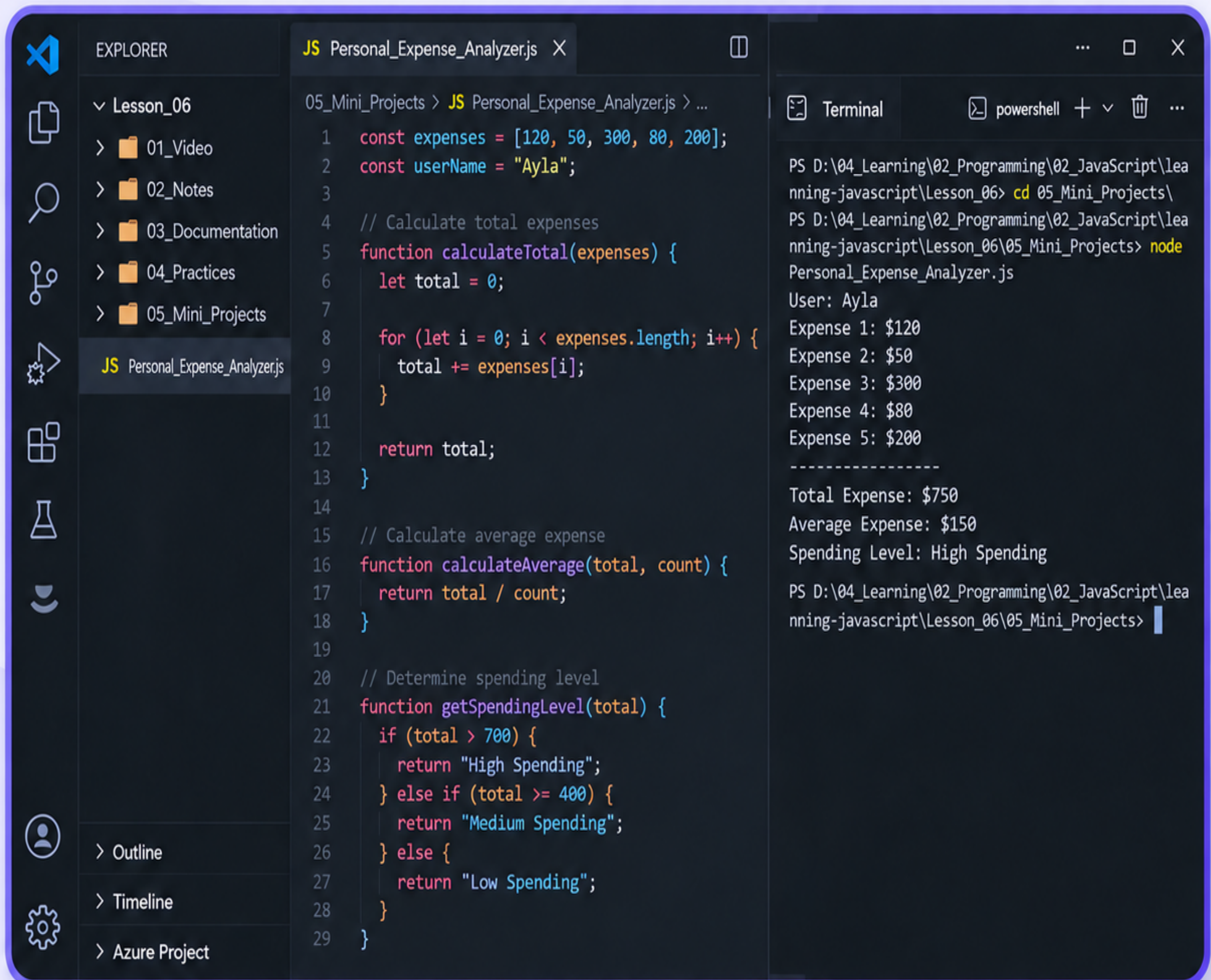
- Numbers store numeric values.
- Strings store text values.
- typeof checks the data type.
- Methods transform and format values.
- Explicit conversion helps avoid bugs.



JS

JavaScript Practice Output

Lesson 06 • Practice 01



The screenshot displays the Visual Studio Code interface with a project named 'Lesson_06'. The Explorer sidebar on the left shows the file structure, including folders for '01_Video', '02_Notes', '03_Documentation', '04_Practices', and '05_Mini_Projects'. The '05_Mini_Projects' folder is expanded, showing the file 'JS Personal_Expense_Analyzer.js'. The main editor window displays the code for 'Personal_Expense_Analyzer.js', which includes an array of expenses, a function to calculate the total, a function to calculate the average, and a function to determine the spending level. The terminal window on the right shows the command 'node Personal_Expense_Analyzer.js' being executed, resulting in the following output:

```
PS D:\04_Learning\02_Programming\02_JavaScript\learning-javascript\Lesson_06> cd 05_Mini_Projects\
PS D:\04_Learning\02_Programming\02_JavaScript\learning-javascript\Lesson_06\05_Mini_Projects> node
Personal_Expense_Analyzer.js
User: Ayla
Expense 1: $120
Expense 2: $50
Expense 3: $300
Expense 4: $80
Expense 5: $200
-----
Total Expense: $750
Average Expense: $150
Spending Level: High Spending
PS D:\04_Learning\02_Programming\02_JavaScript\learning-javascript\Lesson_06\05_Mini_Projects>
```



Lesson 06

Practice 01

JavaScript